

Algorithmic Market Making Case

Rotman International Trading Competition 2026

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OVERVIEW

The Algorithmic Market Making Case is designed to challenge participants' programming skills by developing algorithms using the RIT API to automate trading strategies and react to changing market conditions driven by news releases. Throughout the case, these algorithms will need to provide liquidity while managing risk from sudden price movements. Participants must balance profit-seeking market making activities with strict position limits. Due to the high-frequency nature of the case, participants are encouraged to develop algorithms that can adapt to rapid changes in market dynamics using their selected programming languages.

DESCRIPTION

There will be 12 heats with 1 team member competing in each heat. Only one team member shall trade to represent the team for all heats. Each heat will be 5 minutes long and represent one trading week (5 trading days), with each minute representing one full trading day.

Parameter	Value
Number of trading heats	12
Trading time per heat	5 minutes (300 seconds)
Calendar time per heat	One trading week (5 days)

Order submission using the RIT API will be enabled. Data retrieval via Real-time Data (RTD) Links and the RIT API will also be enabled. **All trades must be executed by a trading algorithm.** Participants will not be allowed to trade manually through the RIT User Application once the heat begins. Participants are allowed to modify their algorithms in response to prevailing market conditions and competition from the algorithms of other teams. They will have 2 minutes between each heat to re-load their algorithms. A base template algorithm will be provided to participants and can be directly modified for use in the competition. However, participants are strongly encouraged to create their own algorithms.

MARKET DYNAMICS

This case involves four stocks with the following details.

Ticker	SPNG	SMMR	ATMN	WNTR
Security type	Stock	Stock	Stock	Stock
Quote currency	CAD	CAD	CAD	CAD
Starting Price	\$25	\$25	\$25	\$25
Fee/share (Market orders)	\$0.02	\$0.02	\$0.02	\$0.02

Rebate/share (Limit/Passive orders)	\$0.01	\$0.02	\$0.015	\$0.025
Max order size	10,000	10,000	10,000	10,000

Your desk is market-making in four stocks, and you expect a volatile week. After each market close (i.e., every minute), your team anticipates a potential news release that may affect these stocks. While the news will not be provided to you directly, your analyst team has indicated that these releases will be the primary driver of price movements throughout the week.

The magnitude and direction of each stock's reaction to a given news event may differ from minute to minute. Some stocks may react more strongly than others, and correlations among stock movements may vary over time.

Given this potential volatility, the Chief Risk Officer (CRO) has imposed an aggregate position limit to control overnight market risk exposure. This limit will be announced at the beginning of the week (i.e., at the start of each heat) and is calculated based on your aggregate position across all four stocks:

Aggregate Position Limit

$$= |Position_{SPNG}| + |Position_{SMMR}| + |Position_{ATMN}| + |Position_{WNTR}|$$

Penalty for Exceeding Limit: \$10 per share for every share above the limit, assessed immediately at each market close (i.e., every minute).

Example: If your aggregate position limit is 15,000 shares and you hold:

- Long 12,000 shares of SPNG
- Short 3,000 shares of SMMR
- Long 7,000 shares of ATMN
- Long 6,000 shares of WNTR

Your aggregate position = $|12,000| + |-3,000| + |7,000| + |6,000| = 28,000$ shares

You are 13,000 shares over the limit, incurring a penalty of $13,000 \times \$10 = \$130,000$

This penalty structure encourages aggressive market making (providing liquidity on both sides) while penalizing excessive directional risk-taking and suboptimal position management.

TRADING/POSITION LIMITS AND TRANSACTION COSTS

In addition to the aggregate position limit for every market close, each trader will be subject to gross and net trading/position limits during trading in each heat. The gross limit reflects the sum of the absolute values of the long and short positions across all securities, and the net limit reflects the sum of long and short positions such that short positions negate any long positions. Anyone who exceeds the trading/position limits will incur a penalty of \$5 per share.

The maximum trade size will be 10,000 shares per order for each stock. Transaction fees will be set at \$0.02 per share for each security on all market orders filled. A rebate will be provided for all submitted limit orders that are filled.

POSITION CLOSE-OUT

Any non-zero position of stocks will be closed out at the end of trading based on the last traded price for each stock.

KEY OBJECTIVES

Objective 1

Develop an algorithmic market making strategy that continuously quotes bid and ask prices to profit from the bid-ask spread while managing inventory risk. Your algorithm should provide liquidity to the market while avoiding excessive position accumulation.

Objective 2

Manage total exposure across all four stocks, potentially by flattening positions as they approach the Limit, balancing long and short positions across different stocks, and implementing risk controls that prevent excessive position accumulation.